Smoke Rescue

Two quadruped robots search a burning, smoke-filled house with OpenCV 5 perception, live visual SLAM between the robots, and agents that act only on cited OpenCV evidence. OpenCV AI Competition 2026 technical report, 25 October 2026.
Team: names and roles to be filled in by the team. Repository branch `overnight-25oct`.

1. Problem

A search team in a burning house loses sight to smoke and time to heat, and two searchers who cannot share what they saw search the same rooms twice. We built a simulation of that search with the constraints a real robot team would have: each robot sees only through its own RGB-D camera, thermal camera and planar LiDAR; nobody gets a floor plan; the radio is lossy and short; and every result is scored against ground truth that no robot-side code can read.

The system has to answer three questions live: where am I and where is my partner (visual SLAM), where have we already looked (shared room records over the radio), and is that a person (OpenCV 5 dnn with YOLO11m, never a heat blob alone).

2. Architecture

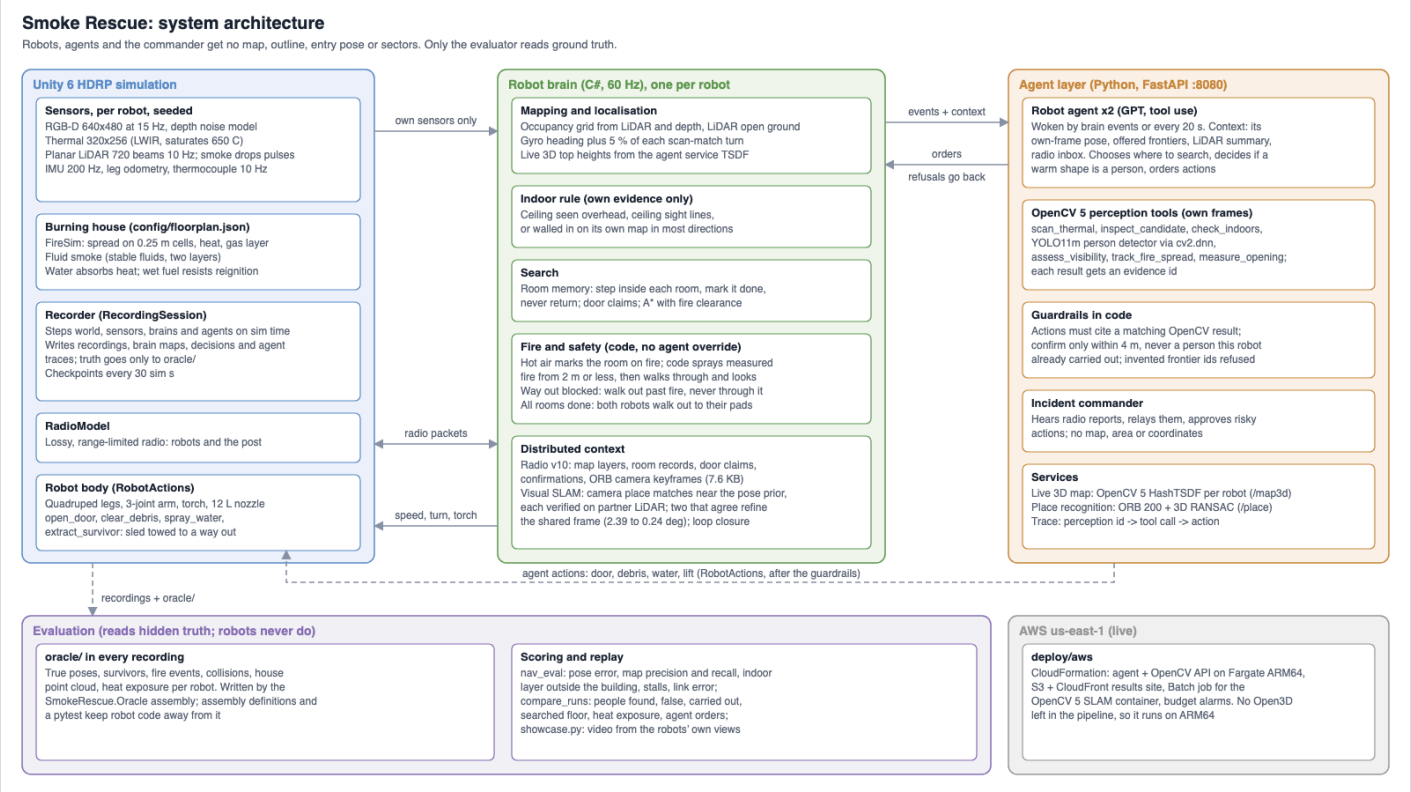


Figure 1. Unity 6 HDRP simulates the house, fire, smoke and sensors. A C# brain per robot maps, plans and keeps the robot safe at 60 Hz. A Python service (FastAPI) runs the OpenCV 5 perception, place recognition and the 3D map, and hosts the GPT agents and the incident commander. The same service runs on AWS Fargate. Only the evaluator reads ground truth (Unity `SmokeRescue.Oracle` assembly and `oracle/` in each recording).

3. Where OpenCV 5 does the work

Component	OpenCV 5 API	What it decides	Measured
Person detector	`cv2.dnn`, YOLO11m ONNX, range from own depth	Only a YOLO detection confirms a person; a thermal shape alone never does	View recall 0.443, 0.024 false boxes per empty frame at conf 0.25, over 1212 recorded frames (729 person views, 500 negatives); 1266 ms per frame on the M5 CPU
Place recognition (visual SLAM front end)	ORB (3000 detected, 200 kept with depth), `BFMatcher` Hamming kNN, ratio test 0.85, RANSAC rigid fit on 3D feature positions	Which partner camera view shows the same place as one of mine, and how our frames sit	200 features: cross-robot recall 11.7 %, same-robot 50 %, 0 false accepts in about 30,000 negative pairs (largest negative 8 inliers, threshold 10); payload 7,605 bytes
3D map	`cv2.Volume` HashTSDF per robot	Obstacle top heights for the planner	Live per robot; exported as `map3d.ply`
Agent perception tools	Thresholding, contours, morphology, Laplacian variance, dark channel on the robot's own frames	`scan_thermal`, `inspect_candidate`, `check_indoors`, `assess_visibility`, `track_fire_spread`, `measure_opening`: every result gets an evidence id the agent must cite	`assess_visibility` 8.6 ms on AWS Fargate ARM64
Rejected	`cv2.Odometry` as the pose source	none	Diverged 17 degrees in 4 s on our frames; the brain uses gyro plus LiDAR scan matching instead

4. Live visual SLAM between two robots

Both robots launch from pads at the one front door. The pads give a shared frame, but a pad placement error turns a robot's whole map: in seed 934 robot B's true heading was 2.39 degrees off its pad. Over a 15 m house that misplaces B's far walls by about 0.6 m, enough to put a door in a wall.

Each robot stores ORB keyframes with depth and broadcasts them over the radio (7.6 KB each, under the 8 KB interface limit). A robot compares a partner keyframe only with its own views near where the shared-frame prior says the partner stood (3 m, 60 degrees). A match must pass RANSAC with 10 or more inliers. It is then checked a second way: the partner's LiDAR sweep from that moment is placed on the robot's own LiDAR map. Two matches that agree within 1 degree and 0.15 m, taken at least 1 m apart, refine the link, bounded to 5 degrees and 0.5 m. The same matcher against the robot's own views 20 s or older measures drift (loop closure) and applies a correction only if the LiDAR fit does not get worse.

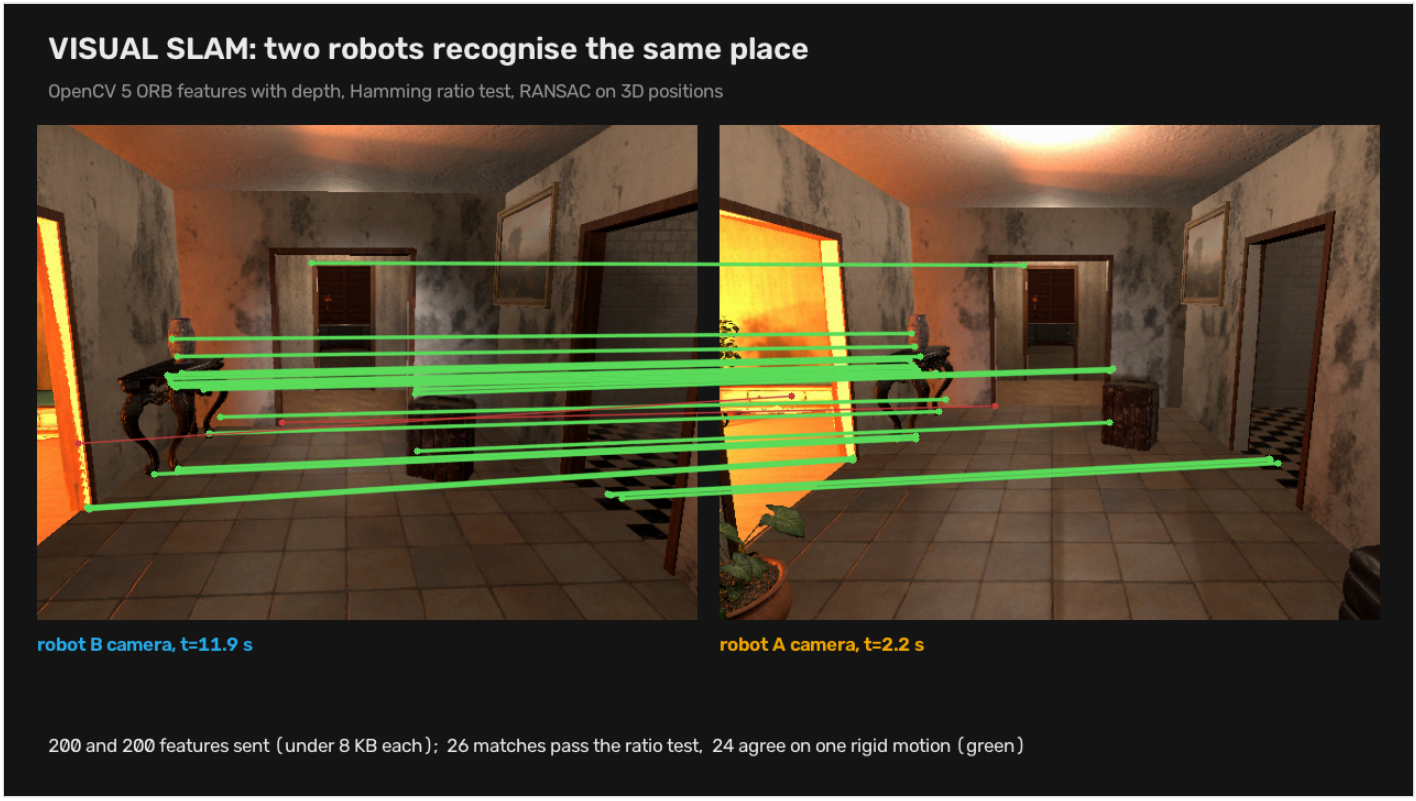


Figure 2. Robot B at 11.9 s and robot A at 2.2 s see the foyer. 26 matches pass the ratio test; 24 agree on one rigid motion (green). Recomputed offline with the robot-side code from the recorded frames; the live event had 32 inliers.

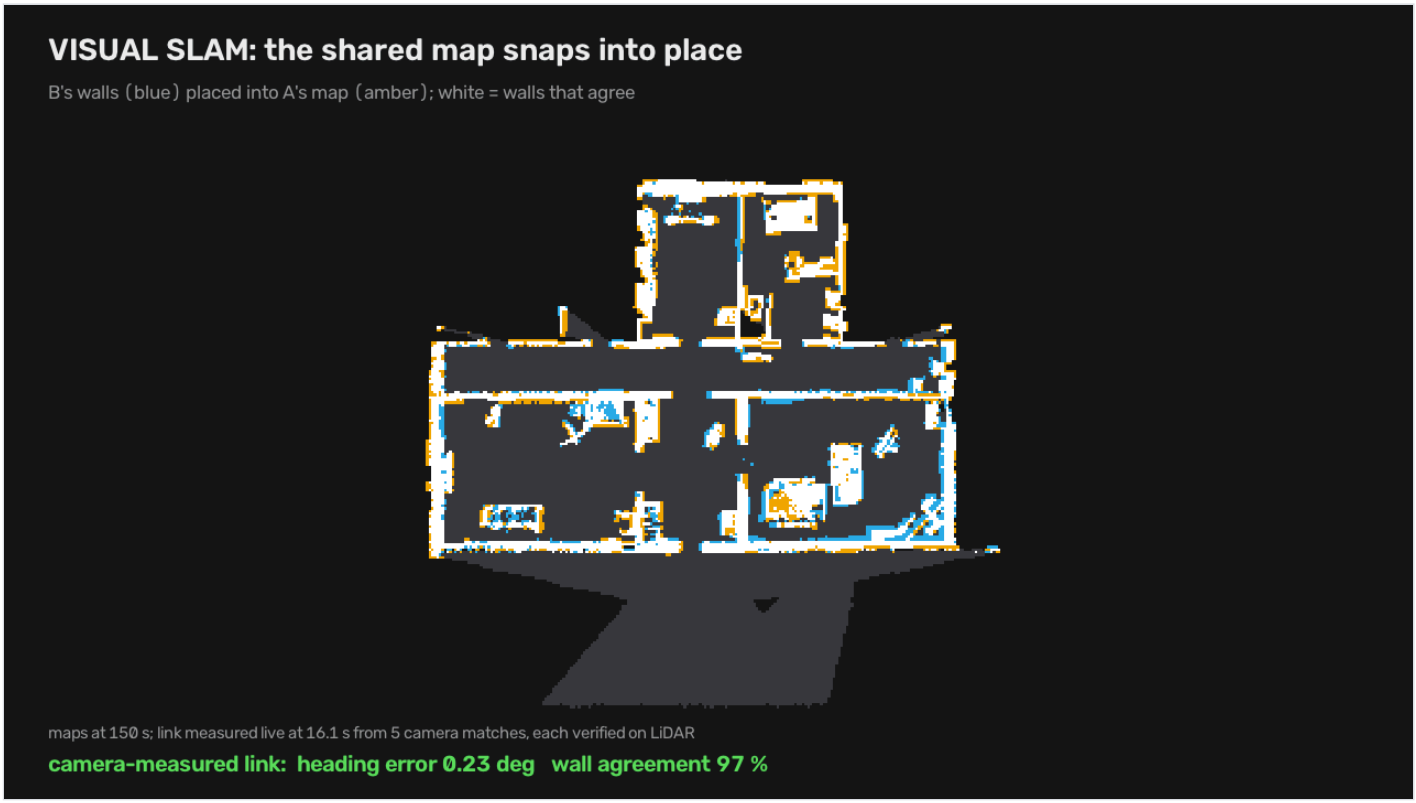


Figure 3. Robot B's walls (blue) placed on robot A's map (amber) through the camera-measured link; white marks walls that agree. Agreement within one grid cell: 72 % with the launch-pad prior, 97 % with the link, and 52 % with the link reversed, which confirms the direction of the transform. Video segment: `tools/visual_slam_clip.py`.

Live run (seed 934)	Link error before	After refine	Camera matches verified	Loop closures (median drift)
vslam1, 24 Oct 23:36	2.39 deg	A 0.91, B 0.30 deg		
allfixes, 25 Oct 00:07	2.39 deg	A 0.02, B 0.30 deg		
full mission, 25 Oct 03:18	2.39 deg	A 0.235 at 16.1 s, B 0.12 deg	A 4	138 total; A median 3.1 cm, 0.25 deg; 0 needed a correction

Limit, stated plainly: cross-robot recall per keyframe pair is 11.7 %. It works here because each robot sends many keyframes and only two agreeing matches are needed. Shared views happen at the entrance; robots that enter by different doors and never share a view would not link until they do.

5. Autonomy: brain, shared context and agents

Robot brain (C#, 60 Hz, one per robot)

Occupancy grid from LiDAR and depth; gyro heading corrected by LiDAR scan matching. Room memory: a robot steps inside each room, marks it searched, and never returns to it. Door claims stop both robots taking the same doorway. Fire is handled in code with no agent override: hot air marks a room on fire; the brain sprays measured fire from 2 m or less (it used to fire from up to 4.6 m and the water landed short), then walks through and looks. When nothing reachable is left, both robots walk back out to their pads. Every constant has a written reason in `config/brain.json`, and no constant describes a particular house.

Distributed context (radio v10)

Map layers, room records, door claims, person confirmations and ORB keyframes travel over a lossy, range-limited radio. Partner data is converted through the measured frame link, never through ground truth.

Agents (GPT via Azure, tool use)

One agent per robot is woken by brain events (a survivor candidate, no frontier left, a goal reached) or every 20 s. It sees only its robot's own pose, offered frontiers, LiDAR summary and radio inbox. It calls the OpenCV tools above and must cite a matching evidence id to act. Python guardrails then check the evidence type and position, allow a confirmation only within 4 m, and refuse invented frontier ids. The C# brain applies a second check (indoors by its own evidence, heat, already leaving) and the refusal reason goes back to the model. An incident commander relays radio reports and approves risky actions. Every step is in `agent_trace.jsonl` and the robot's `decisions.jsonl`.

Measured agent behaviour (all 34 recorded agent runs)

Measure	Value
Agent steps / tool calls	1,628 / 6,403
OpenCV 5 perception calls	2,727 (scan_thermal 937, inspect_candidate 754, assess_visibility 509, track_fire_spread 292, check_indoors 229, measure_opening 6)
Robot orders citing an evidence id	990 of 1,571 (63 %); the rest are orders that need none (turn, team message)
Cited ids produced by a recorded OpenCV call	824 of 863
Guardrail refusals	169: explore 48, approach 46, confirm 42, spray 11, other 22

The commonest refusals: a confirmation from beyond 4 m (29; the agent then approaches and inspects again), a warm region the cited inspection measured as a flat surface (20), exploring during a heat event (14), and an evidence id that names no OpenCV result from that step (7). Example from `unity_house_agents_run19`: at 18.0 s robot A's `explore_frontier` was refused during a heat event; at 35.8 s it ordered `approach_candidate` citing a new OpenCV result. Reproduce with `python3 tools/agent_trace_stats.py data/recordings`.

6. Cloud: AWS

CloudFormation stacks in us-east-1 (`deploy/aws`): the agent and OpenCV 5 API on ECS Fargate ARM64 (1 vCPU, 2 GB) behind an ALB with a WAF rate limit of 2,000 requests per 5 minutes per IP; a Batch job for the offline SLAM container; and budget alarms. The account was not yet verified for CloudFront on 25 October, so the read-only OpenCV routes (`/health`, `/vision/*`, `/perceive/people`, `/place/*/match`) are open on the ALB. The model routes still require a secret origin header and return 403 without it, so nobody outside can spend the model budget.

```
./deploy/aws/smoke_test.sh http://smoke--LoadB-m7oFRuQ63UoE-1386562850.us-east-1.elb.amazonaws.com --direct
PASS GET /health (HTTP 200)              opencv 5.0.0
PASS POST /vision/assess_visibility      8.6 ms
```

7. Evaluation, including failures

Headline numbers, all runs, not the best one. Across all 10 live Unity runs of the seed 934 house the robots confirmed 32 of 50 people (64 %, Wilson 95 % interval 50 to 76 %), with one false confirmation in 10 runs (since removed: thermal-only confirmation is off). Median rooms entered 6 of 10, median floor seen 74 % (n = 9 runs with the true-floor measure). On two houses the brain was never tuned on (headless, below) both robots entered every room and found 5 of 5 people with no false confirmation. The best single run (8 of 10 rooms, 93 %, 5 of 5) is listed with the rest, not instead of them.

The evaluator scores each recording against hidden ground truth: rooms entered, floor seen by a camera, people confirmed (within 1.5 m of a real person) and false confirmations. Ten live Unity runs of the seed 934 house, in order, as the brain changed:

Run	Rooms entered	Floor seen	People	False
23 Oct 23:55	4/10	0 %*	2/5	0
24 Oct 02:16	5/10	70 %	4/5	0
24 Oct 04:53	5/10	71 %	3/5	0
24 Oct 16:35	4/10	69 %	3/5	0
24 Oct 19:54	7/10	81 %	3/5	0
24 Oct 20:06	7/10	74 %	2/5	1
25 Oct 00:07	6/10	90 %	4/5	0
25 Oct 01:41	6/10	70 %	3/5	0
25 Oct 02:36	7/10	74 %	3/5	0
25 Oct 03:18, full mission (best)	8/10	93 %	5/5	0

* Floor measured before the evaluator used the true floor. The one false confirmation (20:06) came from a thermal dwell; thermal-only confirmation was then switched off. The 03:18 run is our best, not our typical: the median of the last five runs is 7 rooms, 74 % floor, 3 of 5 people.

Two houses the robots were never tuned on

Codex built two more one-floor houses from scratch on 25 October: a family ranch (9 rooms, 165 m2) and a long ranch (11 spaces on a central hallway, 198 m2), each with one front door, five people and a bedroom fire. Nothing in the brain changed for them. Headless full missions (ray-cast sensors, fire-driven air, the same brain and radio; no rendered camera):

House	Rooms entered	Floor seen	People	False	Both robots out
Family ranch (final brain, 25 Oct 15:30)	9/9	98.2 %	5/5	0	yes, 516 and 545 s
Family ranch (Codex run)	9/9	97.9 %	5/5	0	yes
Long ranch (Codex run)	11/11	95.9 %	5/5	0	yes

Headless results are not live Unity results: the headless simulator has no rendered camera, so person confirmation there does not exercise YOLO. Live Unity missions with the Terra agents on both houses are recorded separately.

Where it fails

- **Until 25 October no live run ended with both robots outside.** Three causes, each found in a recording and each fixed with a regression test that fails without the fix: (1) once inside, a robot locks the floor at its way in, and the end-of-search walk out planned against that lock, so no route existed at any fire margin; (2) room memory skips rooms the partner has claimed, so a robot with nothing of its own left walked out at 143 s while the team still had unentered rooms (it now waits until no remembered room is unentered by either robot); (3) the robot that finished first stopped on its pad in the front doorway, and its partner's map then showed a body across the only door (it now steps 1.5 m out and 1.5 m aside). Logic suite 279/279. Live confirmation: run in progress.
- Under memory pressure the place service timed out (68 and 69 "Operation aborted" errors for robots A and B in the 25 Oct 11:25 run), and that run found 4 of 5 people with 82 % floor. Perception latency depends on the host.
- Person recall per camera view is 0.443 (YOLO11m on our rendered frames): people seen from far, from behind furniture or through smoke are often missed on a single view, and the robot must get closer.
- Cross-robot place recognition accepts 11.7 % of true same-place keyframe pairs; the link works because many keyframes are sent and only two agreeing matches are needed. Robots that never share a view do not link.
- Room memory measured headless on 12 seeded houses: 79/110 rooms entered without it, 80/110 with it, 82/110 with door claims added; floor 76.8, 75.4 and 78.6 %. A small gain, not a solved problem.
- Recording is slow: about 5 simulated seconds per wall minute with live perception, so each full mission takes about two hours.

8. Reproduce

```
# Python 3.11, pinned in brain/uv.lock (opencv-python 5.0.0)
cd brain && uv sync && uv run pytest -q              # 532 tests
# C# brain logic without Unity (.NET from Unity 6000.6.6f1)
./tools/headless/test.sh                               # 274 passed, 0 failed
# Live: Unity 6000.6.6f1 HDRP, scene Assets/SmokeRescue/Scenes/House.unity
#   Tools > Smoke Rescue > Record > House one-door seed 934 full mission
cd brain && uv run uvicorn brain.agent.server:app --port 8080   # OpenCV 5 service
# Score a recording against ground truth
brain/.venv/bin/python tools/run_audit.py data/recordings/<run>
brain/.venv/bin/python tools/visual_slam_eval.py data/recordings/<run>
# AWS
deploy/aws/deploy.sh --email <you> ...  then  deploy/aws/smoke_test.sh <url> --direct
```

9. Responsible operation

- Robots never read ground truth: Unity assembly definitions and a pytest check keep robot code away from the oracle.
- Agents cannot act without OpenCV evidence; risky actions need commander approval; refusals are logged and fed back.
- Cloud spend is capped by budget alarms, the WAF rate limit, and model routes that are closed to the public.
- This is a simulation. The fire model is an approximation for demonstration, not a validated fire-engineering solver (`config/fire.json` says so). No result here claims real-world rescue performance.